MTH501

QUIZ-3

SOLVED BY JUNAID MALIK



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ALL answers are verified if found any mistake then Correct ACCORDINGLY

Q 1 :A partitioned square matrix 'A' is said to the matrix if the matrices on the main diagonal square and all matrices above the main diagonal are zero.

- Block lower triangular

Q2: if there is a vector v= (2,1,0) then ||v|| is ……..

- $\sqrt{5}$

Q3: Each pair of eigenvalue and its corresponding eigenvectors provide a solution of equation.

Eigen function

Q4: which of the following is true for the matrix

Where M_{11} ,M_{22} and M_{33} are square sub – matrices, and O is Zero sub matrix?

It is block upper triangular matrix

Q5: suppose that real solution y_1 and y_2 of $x^1 = Ax$ from a basis for the two- dimensional real vector space if y_1 and y_2 are ……..

Linearly independent

Q6: Let W be a subspace of R and ($u_1, u_2,.....u_p$) is any orthogonal basis of W, then $y = c_1u_1 + c_2u_2 \ldots\ldots\ldots c_nu_n$ where

$$Cj = \frac{y.uj}{uj.uj}$$

Q7: Elementary row operation on a matrix do not affect the ….. relation among the column of the matrix

Linear dependence

Q8: Two vector are …….. if at least one of the vector is a multiple of the other

Linearly dependent

Q9: Electric circuit rotation is caused by sine and cosine function if there eigenvalues are complex and hence the origin is called….

Spiral point

Q10: Multiplication of a partitioned matrix by a scalar is also computed…..

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Column by column

Q11: The invariable Matrix .Theorem applies only to……

Square

Q12:Let A be a real 2 by 2 matrix with complex eign values ∂ = a – b + I,(b $\neq$) and associated eigenvectors v in C^2 then A PCP $^{-1}$ where p= ------

P = [Revlmv]

Q13: Let Ax = 0 be a homogeneous linear system of 'n' unkown , then the cowfficient matrix 'A' is invertable if and only if this system has ---------- solution

Trivial

Q14: the two vector are said to b equivalent if

Same length and same direction

Q15: if one of the eigenvalues of $[A]_{nxn}$ is zero .it implies

The determined of [A] is zero

Q16: let be a five – dimensional vector space. And let ba subset of consisting of five vectors.

Mast be linearly dependent , but may or may not span V

Q17: Supposex,y ,z are some vectors in an inner producct space (v,<,>) such that <X,Y>=<X,Z> for all x $\sum$ V then Y=Z

TRUE

Q18: let U,V and W be vectros in R^n ,then

(U+ v), W=U.W+V.W

Q19: if u and v non zero vector in either R^2orR^3 then by the law of cosines || u – vii^2……..

$||u - v||^2 = ||u||^2 + ||v||^2 - 2\ ||u||||v||\cos$

Q20: If u +v – w, then

V = w

Q21: Two vectors u and v in R^n are orthogonal if…….

u.v = o

Q22:A vector whose length 1 is called ……..

Unit vector

Q23: A ……. Matrix whit orthogonal columns is an orthogonal matrix.

Square

Q24: if a square matrix has orthogonal columns then it also has …… rows.

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Orthogonal

Q25: The norm of v is the non-negative scalar ||v|| defined by

$||v|| = \sqrt{V}\,^2{}_1 + v_2{}^2 + \ldots\ldots\ldots\ldots v^2{}_n$

Q26: the matrix A ^ T x A is invertable if and only if the columns of A are linearly independent

True

Q27: let V be a one Eigen vector then conjugate eigen vector is repented by.

V----

Q28: Any finite dimensional inner product space an orthonormal basis.

True

Q 29: the …….. vector is orthogonal to every vector in R^n.

Zero

Q 30: Each pair of eigenvalue and this its corresponding eigenvector provides a solution of the equation X^1 = Ax which is called …….. of the differential equation.

Eigen function

Q31: if we divide a non – zero vector by its length we get a

Unit vector

Q32: A matrix A – (n\time n) has both positive and negative eigenvalues so in this case origin behave as a

Saddle point

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